Listing 8.2 shows how the cosine similarities are computed using the `cosine_similarity` function from scikit-learn. Similarities are calculated for each document against all other documents in the set. The computed matrix of similarities for documents is stored in the `cosine_similarities` variable. Then, in the input loop, the user can select the document to view its similarities to the other documents.

Listing 8.2 `document_vector_similarity` (cosine similarity)

```
cosine_similarities = cosine_similarity(X)      #1

while True:      #2
    selected_document_index = input(f"Enter a document number
↪ (0-{len(documents)-1}) or 'exit' to quit: ").strip()

    if selected_document_index.lower() == 'exit':
        break

    if not selected_document_index.isdigit() or
↪ not 0 <= int(selected_document_index) < len(documents):
        print("Invalid input. Please enter a valid document number.")
        continue

    selected_document_index = int(selected_document_index)    #3

    selected_document_similarities = cosine_similarities[selected_document_index]
#4

# code to plot document similarities omitted
```

#1 Computes the document similarities for all vector pairs
#2 The main input loop
#3 Gets the selected document index to compare with
#4 Extracts the computed similarities against all documents

Figure 8.5 shows the output of running the sample in VS Code (F5 for debugging mode). After you select a document, you'll see the similarities between the various documents in the set. A document will have a cosine similarity of 1 with itself. Note that you won't see a negative similarity because of the TF–IDF vectorization. We'll look later at other, more sophisticated means of measuring semantic similarity.